Lecture 7

Fourier sampling and phase estimation

Objectives. *Lecture 5 sampled the Fourier transform of $\mathbb{Z}_2^n$, and Lecture 6 made circuits a model of computation. This lecture asks:* how can a circuit read an eigenphase? *The quantum Fourier transform of $\mathbb{Z}_N$, $N = 2^n$, is a circuit of $n(n+1)/2$ gates and $\lfloor n/2 \rfloor$ swaps (Theorem 7.2). With it, phase estimation reads an eigenphase φ of a gate: its nearest t-bit approximation with probability at least $4/\pi^2$, and φ itself when $2^t\varphi$ is an integer (Theorem 7.3). It is the mechanism of factoring (Lecture 8).*

By the end of this lecture you will be able to:

- Define the QFT with its sign, prove that it is unitary, and prove the product formula.
- Read the QFT's circuit off the product formula, and count its gates, before and after compiling.
- Prove that phase estimation reads the nearest t-bit approximation of an eigenphase with probability at least $4/\pi^2$, and compute its readings for the phase $\frac{1}{3}$.
- Show that on a superposition of eigenvectors it reads each eigenphase with its weight.

7.1 The QFT and its sign

Let $N = 2^n$. A number $x \in \{0, \dots, N-1\}$ is written in n bits, the top wire first and most significant: $x = \sum_{j=1}^{n} x_j 2^{n-j}$, and $|x\rangle = |x_1\rangle \otimes \cdots \otimes |x_n\rangle$, as in (1.3).

Definition 7.1 (The quantum Fourier transform)**.** With $\omega = e^{2\pi i/N}$, QFT_N is the matrix on n qubits with

$$\mathrm{QFT}_N |x\rangle = N^{-1/2} \sum_y \omega^{xy} |y\rangle \tag{7.1}$$

for every x, the sum over $y = 0, \dots, N-1$.

For $N = 2$ it is H. For $N = 4$, with rows y and columns x,

$$\mathrm{QFT}_4 = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix},$$

which is not $H \otimes H$, whose entries are real. The coefficient $\chi_y(x) = \omega^{xy} = e^{2\pi i xy/N}$ is a character of the group $\mathbb{Z}_N$ of the integers modulo N, $\chi_y(x + x') = \chi_y(x)\chi_y(x')$, with the sign of *Symmetries*, L3. So QFT_N is the Fourier transform of $\mathbb{Z}_N$, as $H^{\otimes n}$ is that of $\mathbb{Z}_2^n$: the same strings, another group. The sign is that of the inverse Fourier transform of *Quantum Mechanics*, $e^{+ipx/\hbar}$. NumPy's `fft` has the other sign; QFT_N is its `ifft` with `norm="ortho"`.

QFT_N is unitary. Columns x and x' have the inner product $N^{-1}\sum_y q^y$ with $q = e^{2\pi i(x'-x)/N}$. This is 1 if $x = x'$; otherwise $q \neq 1 = q^N$, and the geometric sum gives

$$\frac{1}{N}\sum_{y=0}^{N-1} q^y = \frac{q^N-1}{N(q-1)} = 0.$$

For an integer k, the state

$$\mathrm{QFT}_N\,|k\rangle = N^{-1/2}\sum_x e^{2\pi ikx/N}\,|x\rangle \tag{7.2}$$

has amplitudes of one modulus and a phase that grows linearly in x, by $2\pi k/N$ a step: for $N = 4$ and $k = 1$ it is the second column of QFT_4, $(|0\rangle + i\,|1\rangle - |2\rangle - i\,|3\rangle)/2$, a quarter turn a step. QFT_N^{-1} maps it to $|k\rangle$: a reading gives the slope k with certainty. A reading after QFT_N^{-1} is *Fourier sampling* on $\mathbb{Z}_N$. Phase estimation makes such a state with a slope that need not be an integer.

7.2 The product formula, the circuit and its compiled count

For a real θ, let $|+_\theta\rangle = (|0\rangle + e^{2\pi i\theta}\,|1\rangle)/\sqrt{2}$; so $|+_0\rangle = |+\rangle$ and $\left|+_{1/2}\right\rangle = |-\rangle$.

Theorem 7.2 (The QFT circuit). *(i) For every x, with qubit j counted from the top,*

$$\mathrm{QFT}_N\,|x\rangle = \bigotimes_{j=1}^{n}\left|+_{x/2^j}\right\rangle. \tag{7.3}$$

(ii) QFT_N is a circuit of $n(n+1)/2$ gates, n Hadamards and $n(n-1)/2$ gates $\mathrm{C}(R_k)$ with $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$, followed by $\lfloor n/2\rfloor$ swaps. · *sketched*

Proof. (i) With $y = \sum_j y_j 2^{n-j}$, $xy/N = \sum_j xy_j/2^j$. As $|y\rangle = |y_1\rangle \otimes \cdots \otimes |y_n\rangle$,

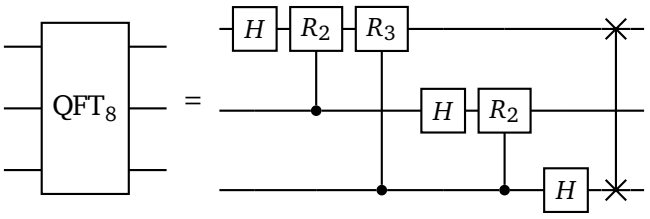
$$\omega^{xy}\,|y\rangle = \bigotimes_j e^{2\pi ixy_j/2^j}\left|y_j\right\rangle.$$

The sum over the strings y is a sum over each $y_j \in \{0,1\}$ on its own wire, and

$$2^{-1/2}\sum_{y_j} e^{2\pi ixy_j/2^j}\left|y_j\right\rangle = \left|+_{x/2^j}\right\rangle.$$

(ii), a sketch. Write $0.b_1\cdots b_m = \sum_i b_i 2^{-i}$. As $x/2^j$ is an integer plus $0.x_{n-j+1}\cdots x_n$, qubit j of (7.3) depends only on the last j bits of x: the top one on x_n, the bottom one on all of them. The circuit builds the factors bottom first. On wire 1, H gives $\left|+_{0.x_1}\right\rangle$, since $(-1)^{x_1} = e^{2\pi ix_1/2}$. Then, for $m = 2,\ldots,n$, $\mathrm{C}(R_m)$ with wire m as its control multiplies the $|1\rangle$ term by $e^{2\pi ix_m/2^m}$; it is diagonal, so the control keeps $|x_m\rangle$. Wire 1 now holds $\left|+_{0.x_1\cdots x_n}\right\rangle$, the factor of qubit n. Wire 2 gets H and the controlled phases from wires $3,\ldots,n$, and holds the factor of qubit $n-1$; and so on, wire k getting one H and $n-k$ controlled phases, $n(n+1)/2$ gates in all. The factors come out in reverse order, and swapping wire k with wire $n+1-k$, for $k \le n/2$, puts them in order. ∎

For $n = 3$,



(7.4)

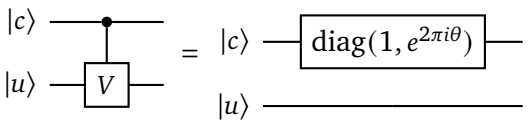
an equality of matrices. On $|001\rangle$, that is $x = 1$, only the phases controlled by wire 3 act. Wire 1 gets H and R_3, and ends in $|+_{1/8}\rangle$; wire 2 gets H and R_2, and ends in $|+_{1/4}\rangle$; wire 3 ends in $H|1\rangle = |+_{1/2}\rangle$. The swap puts $|+_{1/2}\rangle$ on top, and the three factors are those of (7.3) for $x = 1$, in order.

Nielsen and Chuang's Fig. 5.1 (p. 219) leaves the swaps out, so its output is bit-reversed; their Box 5.1 (p. 220) draws the swap. This is not a fast Fourier transform: the N amplitudes stay in the register, and a reading returns one y, with the probability its amplitude gives.

Compiled. The circuit has $L = n(n+1)/2 + \lfloor n/2 \rfloor \le n^2$ gates, each on at most two qubits, and the entries of R_k can be computed to b bits, by their power series, in time polynomial in b and k. So Corollary 6.10 gives, for every $\delta > 0$, a circuit over the alphabet of $O(n^2 \log^c(n^2/\delta))$ gates, written in time polynomial in n and $\log(1/\delta)$, that gives every outcome on every input state with a probability within δ of QFT_N's. The count is cited, as Solovay–Kitaev is.

7.3 Phase estimation

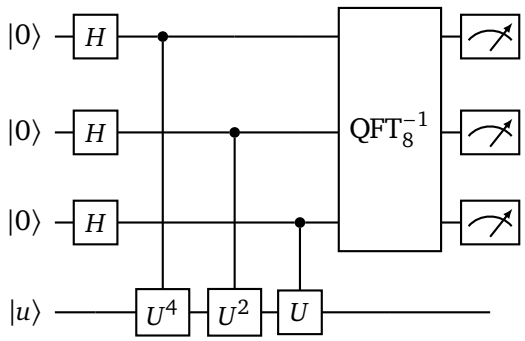
Let U be a gate on m qubits with a unit eigenvector u, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $\varphi \in [0,1)$: its *eigenphase*. Given the state $|u\rangle$ and the gates $\mathrm{C}(U^{2^k})$, we want φ. The tool is Lecture 2's kickback, now for any gate: if $V|u\rangle = e^{2\pi i\theta}|u\rangle$, then



(7.5)

an equality of vectors for every state $c = c_0|0\rangle + c_1|1\rangle$ of the control: $\mathrm{C}(V)(c \otimes u) = c_0|0\rangle \otimes u + c_1 e^{2\pi i\theta}|1\rangle \otimes u$. On $c = |+\rangle$ the control ends in $|+_\theta\rangle$. The phase lands on the control, and the target is unchanged. For $V = X$ and $u = |-\rangle$, with $\theta = \frac{1}{2}$, this is Example 2.8.

The circuit has t counting qubits on top, each through H. Counting qubit j, from the top, controls $U^{2^{t-j}}$; then $\mathrm{QFT}_{2^t}^{-1}$ acts on the counting register, which is read. For $t = 3$:



The reading $y \in \{0, \dots, 2^t - 1\}$ gives the estimate $y/2^t$.

As $U^{2^{t-j}}u = e^{2\pi i 2^{t-j}\varphi}u$, (7.5) on each counting qubit in turn leaves u unchanged and the counting register in

$$\begin{aligned} &\textstyle\bigotimes_{j=1}^{t} |+_{2^{t-j}\varphi}\rangle \\ &\quad = 2^{-t/2} \textstyle\sum_{x=0}^{2^t-1} e^{2\pi i\varphi x}|x\rangle\,, \end{aligned} \tag{7.6}$$

since the term with the bits x_j carries $e^{2\pi i\varphi \sum_j x_j 2^{t-j}} = e^{2\pi i\varphi x}$. This is (7.3) with $n = t$ and $x/2^t$ replaced by φ: a phase growing linearly in x, with the slope $2\pi\varphi$. If $2^t\varphi = k$ is an integer, it is $\mathrm{QFT}_{2^t}|k\rangle$ by (7.2), and the reading is k. Otherwise no reading is exact. The distance of phases is circular,

$$d(\varphi,\theta) = \min_{\ell\in\mathbb{Z}} |\varphi - \theta - \ell|\,,$$

as φ and $\varphi+1$ give one eigenvalue. Some grid point $y/2^t$ is within 2^{-t-1} of φ, and two are when φ is halfway between them.

Theorem 7.3 (Phase estimation). *In this circuit the reading y has the probability $|a_y|^2$, where*

$$a_y = 2^{-t} \sum_{x=0}^{2^t-1} e^{2\pi i x(\varphi - y/2^t)}.$$

(i) If $2^t\varphi$ is an integer, the reading is $2^t\varphi$ with certainty.

(ii) Every y with $d(\varphi, y/2^t) \le 2^{-t-1}$ is read with probability at least $4/\pi^2 \approx 0.405$.

Proof. $\mathrm{QFT}_{2^t}^{-1} = \mathrm{QFT}_{2^t}^{\dagger}$ has the entry $2^{-t/2}e^{-2\pi i xy/2^t}$ in row y and column x. Applied to (7.6), it gives $|y\rangle \otimes u$ the amplitude a_y, and by (3.2) the reading is y with probability $|a_y|^2$. Let $\delta = \varphi - y/2^t - \ell$, with the integer ℓ for which $|\delta| = d(\varphi, y/2^t)$. Since x is an integer, $a_y = 2^{-t}\sum_x q^x$ with $q = e^{2\pi i\delta}$. If $\delta = 0$, then $a_y = 1$: this proves (i), and (ii) when $d = 0$. Otherwise, under (ii), $0 < 2^t|\delta| \le \frac{1}{2}$, so $q \ne 1$, and as $|e^{2\pi i\theta} - 1| = 2|\sin\pi\theta|$,

$$|a_y| = \frac{\left|q^{2^t} - 1\right|}{2^t|q-1|} = \frac{|\sin(\pi 2^t\delta)|}{2^t|\sin(\pi\delta)|}.$$

On $[0, \frac{1}{2}]$, $\sin(\pi z)$ is concave and meets $2z$ at both ends, so $\sin(\pi z) \ge 2z$; and $|\sin(\pi\delta)| \le \pi|\delta|$. So $|a_y| \ge 2\cdot 2^t|\delta|/(2^t\pi|\delta|) = 2/\pi$. ∎

The bound is Watrous's (pp. 212–213), stated with the plain distance $|\varphi - y/2^t|$, which covers no reading when φ is near 1: for $\varphi = \frac{39}{40}$ and $t = 3$ no $y/8$ is within $\frac{1}{16}$ of φ, while the nearest reading in the circular distance, 000, has probability 0.88. The proof above is the same in both.

These are the formulas of the ideal circuit. If its gates are compiled so that every probability moves by at most δ (Theorem 6.2), (ii) holds with $4/\pi^2 - \delta$.

The guard bites. The controlled powers must control U itself. A gate compiled up to a global phase, $e^{2\pi i\alpha}U$, has the eigenphase $\varphi + \alpha$ on u, and under a control that phase is read (Example 6.7): the circuit estimates $\varphi + \alpha$, modulo 1. With $t = 1$, $\mathbb{1}$ reads 0 and $-\mathbb{1}$ reads 1, the phase $\frac{1}{2}$, each with certainty.

The cost. The counting register needs t Hadamards and $\mathrm{QFT}_{2^t}^{-1}$, the mirror image of the circuit of Theorem 7.2 with adjoint boxes (Theorem 2.2): $t(t+1)/2$ gates and $\lfloor t/2 \rfloor$ swaps. The controlled powers cost more. Built from $\mathrm{C}(U)$, as $\mathrm{C}(U)^s = \mathrm{C}(U^s)$, they take $2^{t-1} + \cdots + 2 + 1 = 2^t - 1$ uses of it: each further bit of φ more than doubles the count, unless U^{2^k} has a short circuit of its own. The gate of Lecture 8 has one.

7.4 The phase 1/3

Example 7.4 (Q: the phase $\frac{1}{3}$). Let $U = \mathrm{diag}(1, e^{2\pi i/3})$ and $u = |1\rangle$, so $\varphi = \frac{1}{3} = 0.0101\cdots$ in binary, and let $t = 3$. As $8/3$ is not an integer, no reading is exact. The nearest grid point is $\frac{3}{8} = 0.011$, at the distance $\frac{1}{24}$. With $\delta = -\frac{1}{24}$, the proof's formula gives the reading 011 the probability

$$|a_3|^2 = \frac{\sin^2(\pi/3)}{64\sin^2(\pi/24)} \approx 0.688.$$

The grid point on the other side, $\frac{2}{8} = 0.010$, at the distance $\frac{1}{12}$, is read with probability 0.175. The two neighbours of $\frac{1}{3}$ take 0.863 together, and none of the other six readings has more than $3/64 \approx 0.047$.

So $\frac{1}{3}$ is read as 011 with probability 0.69, well above the floor $4/\pi^2$, and never exactly: $2^t/3$ is an integer for no t. More counting qubits refine the grid; they do not make the reading exact.

7.5 The mixture

What if the input is not an eigenvector? A unitary matrix is normal, so it has an orthonormal basis of eigenvectors (Nielsen and Chuang, Box 2.2, p. 72).

Theorem 7.5 (The mixture). *Let $u_1, \dots, u_{2^m}$ be an orthonormal basis of eigenvectors of U, with $Uu_k = e^{2\pi i \varphi_k} u_k$, and run the circuit on $\psi = \sum_k c_k u_k$. The reading y has the probability $\sum_k |c_k|^2 \, p_k(y)$, where p_k is the distribution of Theorem 7.3 for φ_k. The counting register is in the state $\sum_k |c_k|^2 \, |\Phi_k\rangle \langle\Phi_k|$, where Φ_k is its final state for u_k.*

Proof. The circuit is linear, so it maps $|0 \cdots 0\rangle \otimes \psi$ to $\sum_k c_k \Phi_k \otimes u_k$. By (3.2), y is read with the probability

$$\begin{aligned} &\left\| \textstyle\sum_k c_k \langle y|\Phi_k\rangle \, u_k \right\|^2 \\ &\quad = \textstyle\sum_k |c_k|^2 \, |\langle y|\Phi_k\rangle|^2 \,, \end{aligned}$$

as the u_k are orthonormal: there are no cross terms. In the same way (3.5), with the u_k as the basis of the other wires, gives the reduced state. ∎

The register that holds u_k tells the branches apart, as garbage did in Example 5.2: the circuit acts as if k were drawn first, with probability $|c_k|^2$, and then φ_k estimated. When several u_k share one phase φ, their weights add up to $\|\Pi_\varphi \psi\|^2$, with Π_φ the projector onto that eigenspace, whatever basis is chosen in it (Kitaev, Shen and Vyalyi, pp. 113 and 115). For example, with $U = \mathrm{diag}(e^{2\pi i \cdot 3/8}, e^{2\pi i \cdot 5/8})$, $\psi = (|0\rangle + \sqrt{2}\,|1\rangle)/\sqrt{3}$ and $t = 3$, the reading is 011 with probability $\frac{1}{3}$ and 101 with probability $\frac{2}{3}$, and nothing else. Lecture 8 runs the circuit on a uniform superposition of eigenvectors with the phases s/r, and so estimates s/r for a uniformly random s.

Summary.

- *The QFT* of $\mathbb{Z}_N$ has the characters' sign, $e^{+2\pi i xy/N}$ (Definition 7.1).
- *Its circuit:* $n(n+1)/2$ Hadamards and controlled phases, then $\lfloor n/2 \rfloor$ swaps; compiled, $O(n^2 \log^c(n^2/\delta))$ gates, cited (Theorem 7.2).
- *The main result:* phase estimation reads the nearest t-bit approximation of an eigenphase, in the circular distance, with probability at least $4/\pi^2$, and exactly when $2^t\varphi$ is an integer (Theorem 7.3).
- *Handed on:* on a superposition of eigenvectors it samples the eigenphases with their weights (Theorem 7.5); the controlled powers control U itself.
- *Next:* Lecture 8, factoring.

References

Nielsen and Chuang, Box 2.2, p. 72, and §§5.1–5.2, pp. 217–225. Watrous (2025), pp. 202–215. De Wolf, pp. 29–34. Kissinger and van de Wetering, §7.1.5, pp. 308–309. Kitaev, Shen and Vyalyi, pp. 113, 115 and 124–125. A. Yu. Kitaev, *Quantum measurements and the*

Abelian stabilizer problem, arXiv:quant-ph/9511026 (1995), pp. 16–17. C. M. Dawson and M. A. Nielsen, *The Solovay–Kitaev algorithm*, arXiv:quant-ph/0505030 (2005), p. 4.